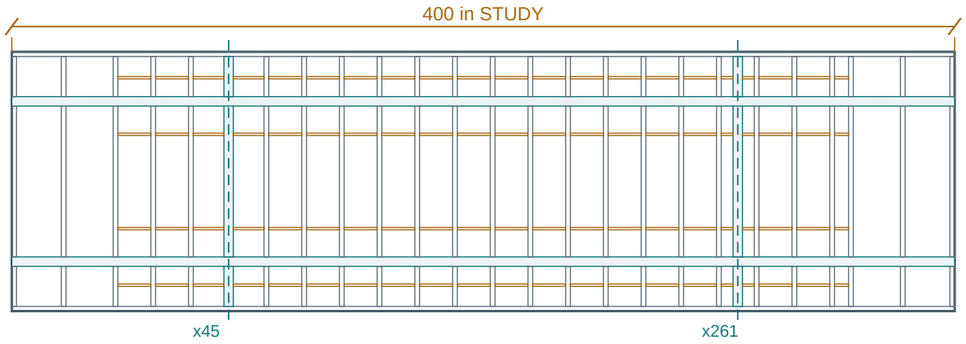
Lighter frame alternatives: retain the usable cabin

Dimensioned member envelopes are coordinated studies. No alternative is selected; no nodes, welds, holes or fabrication tolerances are released.

D0 / 16 in cabin grid / 12 in main girders

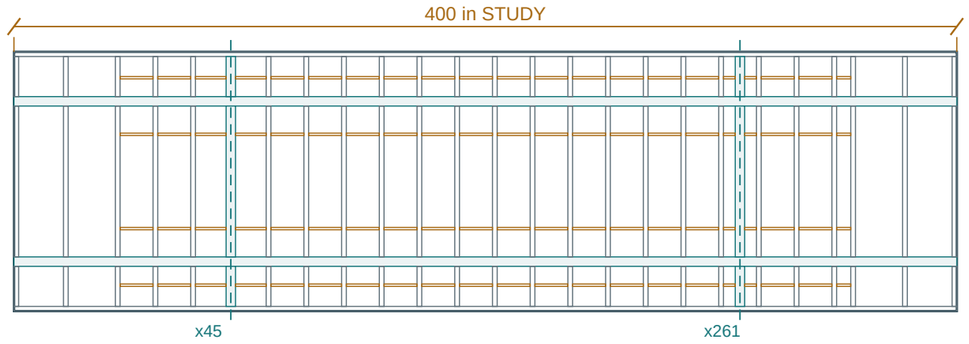
Main 12 x4 x1/4; crossmember + sill 4 x2 x3/16; all bolsters 12 in.



166 members / 1,913.26 kg bare steel; joint and bracket mass excluded.

D1 / 16 in cabin grid / 12 in main girders

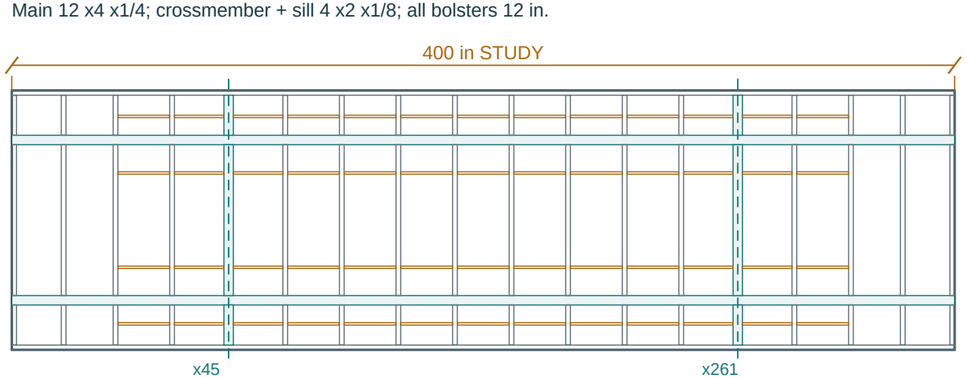
Main 12 x4 x1/4; crossmember + sill 4 x2 x1/8; all bolsters 12 in.



166 members / 1,660.67 kg bare steel; joint and bracket mass excluded.

D2 / 24 in cabin grid / 12 in main girders

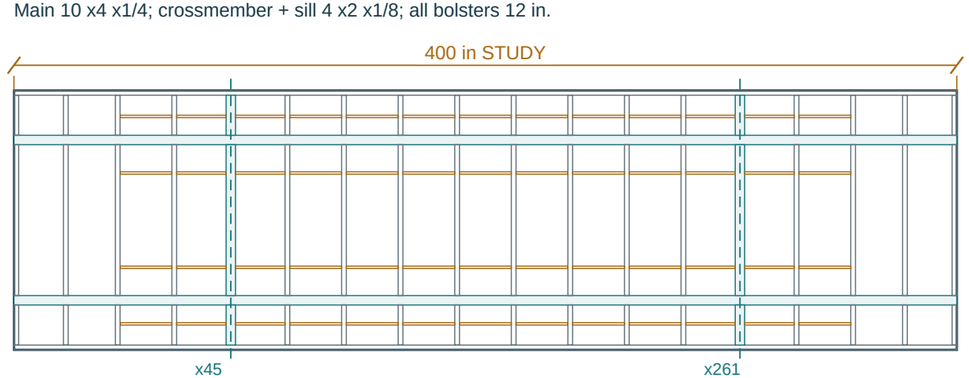
Main 12 x4 x1/4; crossmember + sill 4 x2 x1/8; all bolsters 12 in.



110 members / 1,527.29 kg bare steel; joint and bracket mass excluded.

D3 / 24 in cabin grid / 10 in main girders

Main 10 x4 x1/4; crossmember + sill 4 x2 x1/8; all bolsters 12 in.



110 members / 1,424.47 kg bare steel; joint and bracket mass excluded.

G1 / SELECTION MUST CLOSE ALL FIVE CONDITIONS

Frame mass including joints and brackets; floor-panel response at 24 in support spacing; seat-rail / anchor and local tube-wall response; full frame deflection and fatigue; actual bogie, brake and centre-bolster packaging. A lighter section alone does not establish a viable passenger chassis.

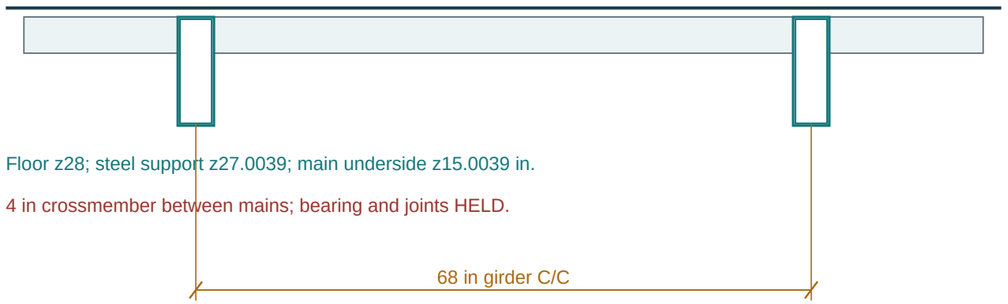
Main and bolster teal; ordinary floor members grey; longitudinal seat rails gold. Model-inch DXFs carry the same alternatives.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Section depth, support nodes and mass reconciliation

Cross-sections use nominal outside dimensions and nominal wall thickness for study drawings; published mass and structural properties govern calculations.

F1 / ORDINARY TRANSVERSE STATION - D0 / D1 / D2



Floor z28; steel support z27.0039; main underside z15.0039 in.

4 in crossmember between mains; bearing and joints HELD.

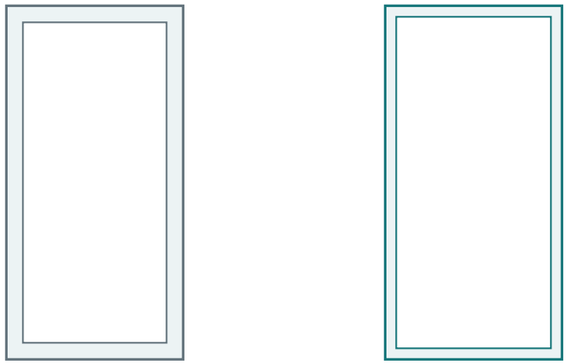
F2 / D3 ORDINARY STATION AND UNCHANGED BOLSTER

10 in main underside z17.0039 in; +2 in locally.

12 in centre bolster still bottoms at z15.0039 in.

D3 does not create 2 in more bogie support height.

F3 / TUBE-WALL AND NODE DETAIL - ENGINEERING LIMITS



D0 nominal 3/16 wall

D1-D3 nominal 1/8 wall

4 x2 floor / sill HSS; sections enlarged for comparison.

F4 / THINNER WALL HOLD

Reduced wall thickness changes local yielding, bearing, punching, welding and fatigue at every anchor and node.

The seat rail remains a load path; it does not replace engineered hard points.

Corner radii and design wall thickness are not depicted here.

F5 / MASS AND STIFFNESS DECISION

Study	Bare steel	Vs 1,500 kg	Saved vs D0
D0	1,913.26 kg	+413.26 kg	0.00 kg
D1	1,660.67 kg	+160.67 kg	252.58 kg
D2	1,527.29 kg	+27.29 kg	385.97 kg
D3	1,424.47 kg	-75.53 kg	488.79 kg

These are member-only takeoffs. Add connections, support seats, brackets, stairs, equipment mounts and coatings before setting tare. D3 is only 75.53 kg below the earlier allowance before those held items.

Seat -> anchor -> rail / floor nodes -> main girder -> bolster -> actual bogie support. Every transfer needs a checked joint. D3 introduces a 2 in depth step at the retained 12 in bolster.

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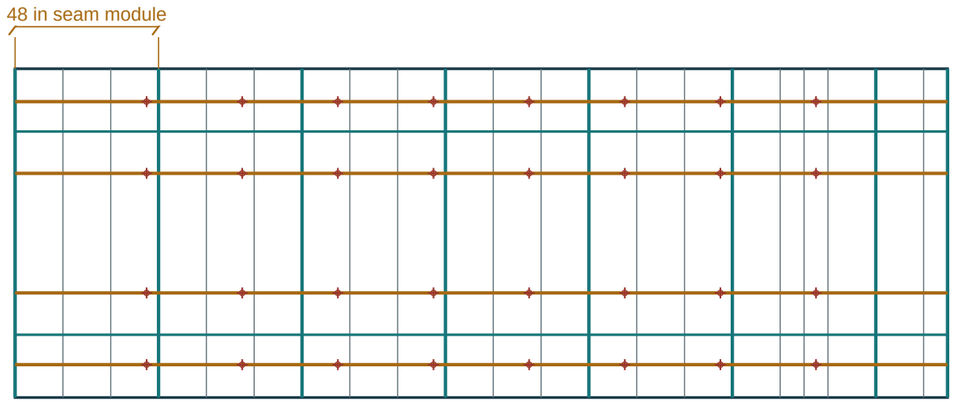
Floor and seating: make the 24 in grid prove itself

The cabin size, eight 32 in seat pitches and aisle are retained. Panel seam support alone is not span approval; seat anchors and floor panels need separate evidence.

P1 / 16 in retained grid

312 x110 in outer floor; y +/-34 in panel-seam support.

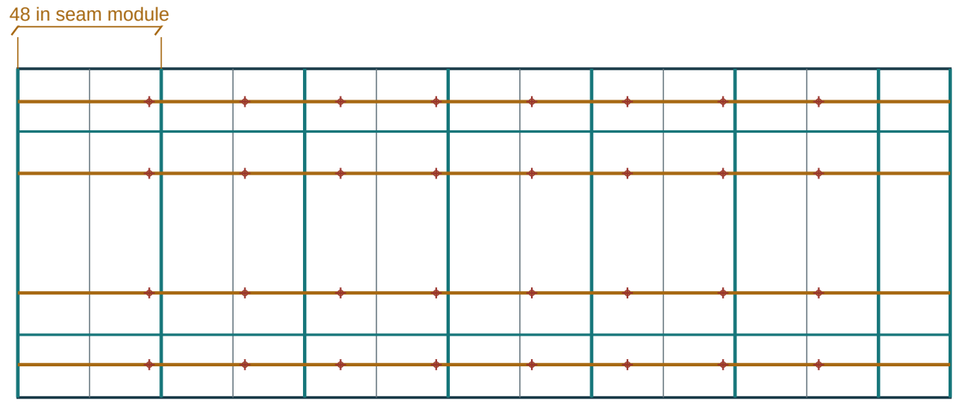


Gold = seat support rails; red = row / rail intersections.

Symbols show load introduction zones; no hole locations released.

P2 / 24 in alternative grid

312 x110 in outer floor; y +/-34 in panel-seam support.



Gold = seat support rails; red = row / rail intersections.

Symbols show load introduction zones; no hole locations released.

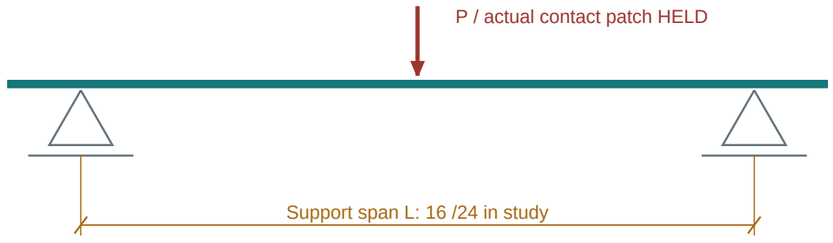
P3 / SAME FLOOR STACK / CHANGE SUPPORT SPAN ONLY

19.05 mm candidate honeycomb panel +2.25 mm candidate finish +1.00 mm assumed adhesive space +3.00 mm assumed support isolation =25.30 mm. Isolation is local to supports. Do not treat the reserve layers as qualified products.

Panel / support span	Deflection proxy	Load trial
3/4 in /16 in	0.343 mm	4.8 kPa
3/4 in /24 in	1.736 mm	4.8 kPa
1 in /24 in	1.076 mm	4.8 kPa

Euler strip proxies use apparent test rigidity; core shear, indentation and creep omitted. Not actual response, allowable load or span approval.

P4 / REPRESENTATIVE PANEL / ANCHOR TEST GEOMETRY



Use actual support, seam, finish and edge-closure assemblies.

Test loads / deflection limits / fatigue cycles: engineer / OEM return.

An intact panel coupon does not release seat anchors or adhesive joints.

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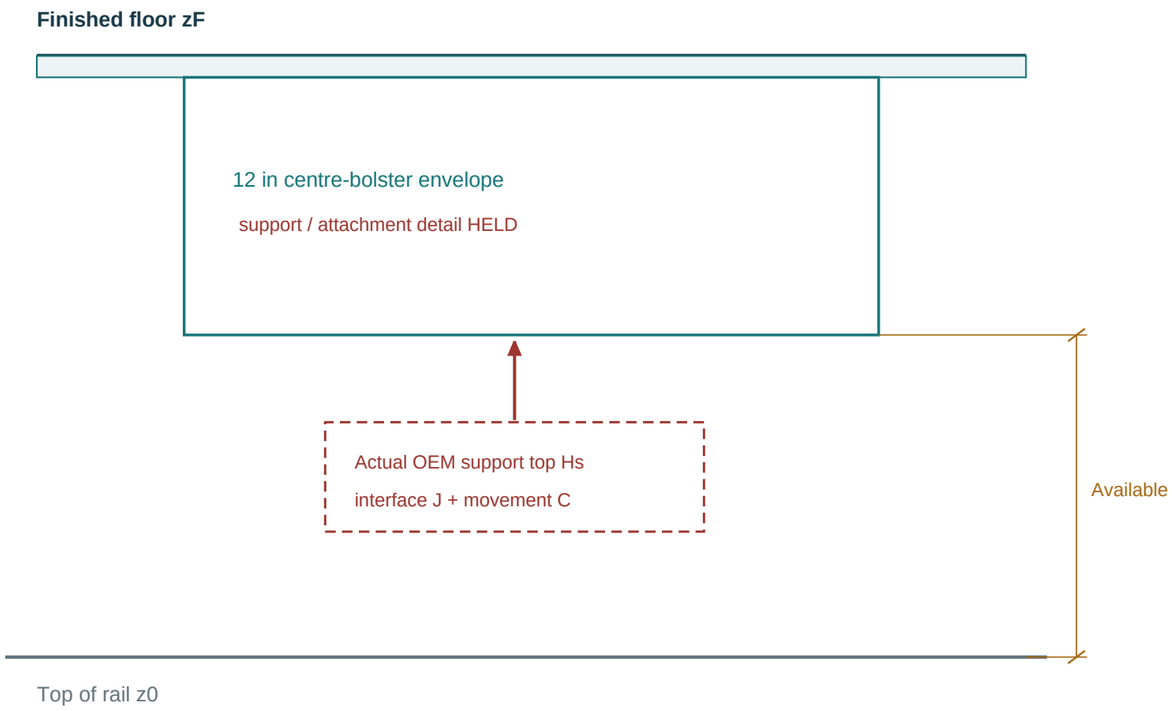
Bogie packaging: floor-height decision and return datums

All actual running-gear support dimensions remain open. A catalog gauge, wheel diameter or load rating alone does not demonstrate fit beneath this coach.

R1 / REQUIRED VERTICAL FIT - SECTION AT CENTRAL BOLSTER



R2 / EXACT GEOMETRIC LIMIT; REAL TERMS UNKNOWN

Hs + J + C <= F -25.30/25.4 -12 [in]

Hs must use the actual loaded/unloaded support datum specified by the bogie engineer. J includes the entire engineered mating stack. C must cover the relative motion and tolerance envelope; no arbitrary safety clearance is inserted.

Floor F	Bolster underside	Status
28 in	15.0039 in	Retained trial
32 in	19.0039 in	Sensitivity
36 in	23.0039 in	Sensitivity

Raising the floor affects every boarding step, bridge/lift, roof height, centre of gravity and clearance. A shallower main girder leaves the retained 12 in centre-bolster limit unchanged.

Separate 1 in panel alternative adds 7.57 kg to the panel takeoff and reduces these available heights by 0.25 in at the same floor. It is not incorporated into D0-D3.

R3 / OEM RETURN MUST SHARE THE DRAWING DATUMS

Return rail-contact datum, new/worn wheel diameter, support top under tare and gross load, pivot interface, side-bearing height and preload, loaded sweep, spring travel, bump/rebound stops, brake-disc envelope, release hardware, maintenance removal path and bounding CAD. State units and identify every supplied interface surface.

No shape or support dimension of an actual bogie is implied by the dashed interface marker.

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Wheel / rail fit and the full swept clearance envelope

36 in gauge is an evaluation basis. Wheel, flange, rail-head, back-to-back and check dimensions must be one matched engineer / supplier system.

W1 / WHEEL-RAIL INTERFACE PARAMETERS - SYMBOLIC SECTION

Dashed wheel/contact zones are not a tread or flange profile.

Actual profiles must provide contact, flange and wear-limit geometry.

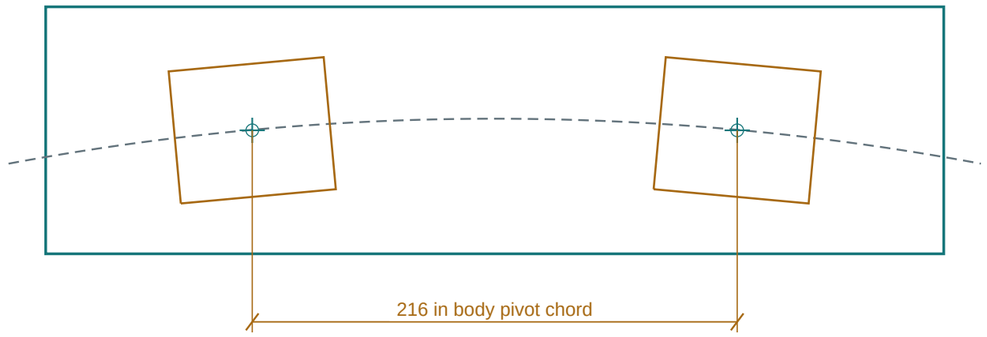
Return: wheel profile + rail section + back-to-back + flange clearance.

Coordinate check rails, turnout flangeways and wear limits as a set.

W2 / PLAN KINEMATICS - BODY AND ROTATING BOGIE

Exact frame box over R30 m curve, at trial 1,165 mm wheelbase.

Gold bogie boxes:1,761 x1,500 mm unselected plan envelopes.



R	Relative yaw	Frame inward	Frame outward
30 m	5.247 deg	1528.4 mm	1675.6 mm
50 m	3.145 deg	1475.7 mm	1569.1 mm
75 m	2.096 deg	1449.4 mm	1513.4 mm

Offsets are from track centreline; zero dynamic / clearance allowance.

W3 / REQUIRED ENVELOPE ADDITIONS

The rigid-body geometry is only the start. Add detailed projections beyond the rectangular frame / platform envelope, including steps and couplings, plus suspension and bearing travel, body roll, track crosslevel and twist, wear, manufacturing tolerance and a reviewed clearance margin. Check reverse curves, adjacent structures, crossings, stations and maintenance routes.

A published 762 mm gauge reference is incompatible with the 914.4 mm evaluation track unless the OEM issues a project-specific engineered variant. No catalog photograph supplies missing interface dimensions.

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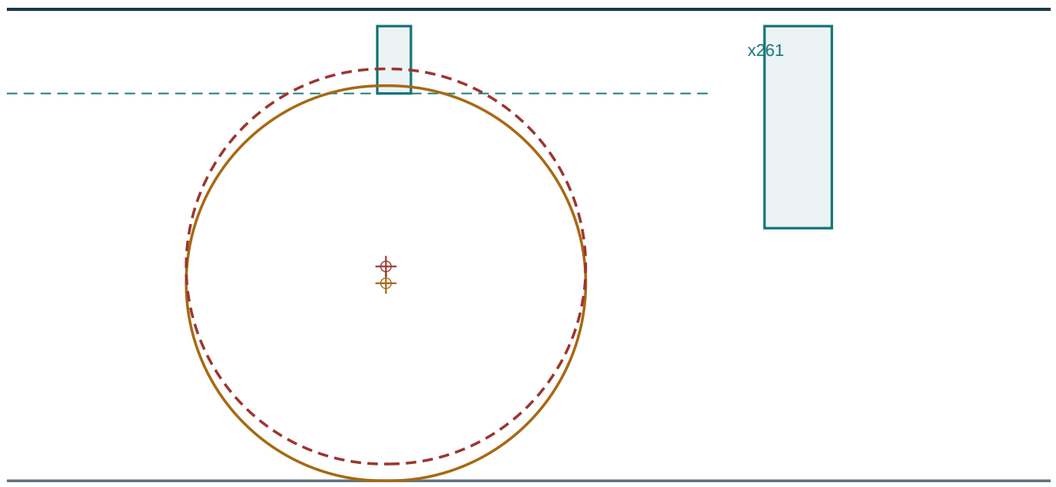
Wheel clearance: a real packaging conflict to resolve

Dimensioned cylinder trials are diagnostic geometry only. They are not a wheel profile, a selected bogie, an acceptance clearance or permission to alter an OEM running-gear assembly.

W4 / SIDE ELEVATION OF 596 mm CYLINDER TRIAL AT x237

Floor28 in / stack25.30 mm / crossmember4 in.

Wheelbase1,165 mm / wheelwidth101.6 mm / gauge contact proxy914.4 mm.



Gold: no heave. Red: +25.4 mm trial relative heave.

Cylinder drawn at -5.247 deg yaw; no flange, play or wear added.

W5 / MEMBER / WHEEL HEIGHT CHECK

Quantity	No heave	+25.4 mm heave
Member underside	584.30 mm	584.30 mm
Wheel upper envelope	596.00 mm	621.40 mm
Geometric gap	-11.70 mm	-37.10 mm

Negative gap indicates an overlap in this trial. The x237 transverse station is retained in all four alternatives. Removing or notching it is not released by this study: it also supports a floor-panel seam.

W6 / SMALLER CYLINDERS - STILL NOT FIT APPROVAL

D / width / wheelbase trial	No heave gap	+25.4 gap
400 /101.6 /1,165 mm	184.30 mm	158.90 mm
500 /101.6 /1,165 mm	84.30 mm	58.90 mm

These gaps concern the specified transverse-member trial only. Bolster, main girders, brakes, axleboxes, suspension and the full actual bogie can govern first.

W7 / REQUIRED DESIGN ACTION

Return the actual 36 in gauge running-gear CAD and matched wheel / rail system. Reconcile wheel diameter, support height, spring travel, brake packaging and the floor height together. Rerun the full three-dimensional sweep with all frame alternatives and real interface parts before altering a member or ordering steel.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Braking force, axle torque and protected power architecture

0% longitudinal grade target retained. Calculations below omit adhesion, response delay and reserve margins; neither a brake selection nor operating limit is established.

H1 / TARGET FORCE AND TORQUE - EQUAL FOUR-AXLE SHARING

Case	Mass	a target	Wheel-rail force
Whole train service	65,100 kg	0.5 m/s2	32.55 kN
Whole train emergency	65,100 kg	0.8 m/s2	52.08 kN
Regular coach emergency	13,200 kg	0.8 m/s2	10.56 kN /2.64 per axle
Accessible emergency	11,000 kg	0.8 m/s2	8.80 kN /2.20 per axle

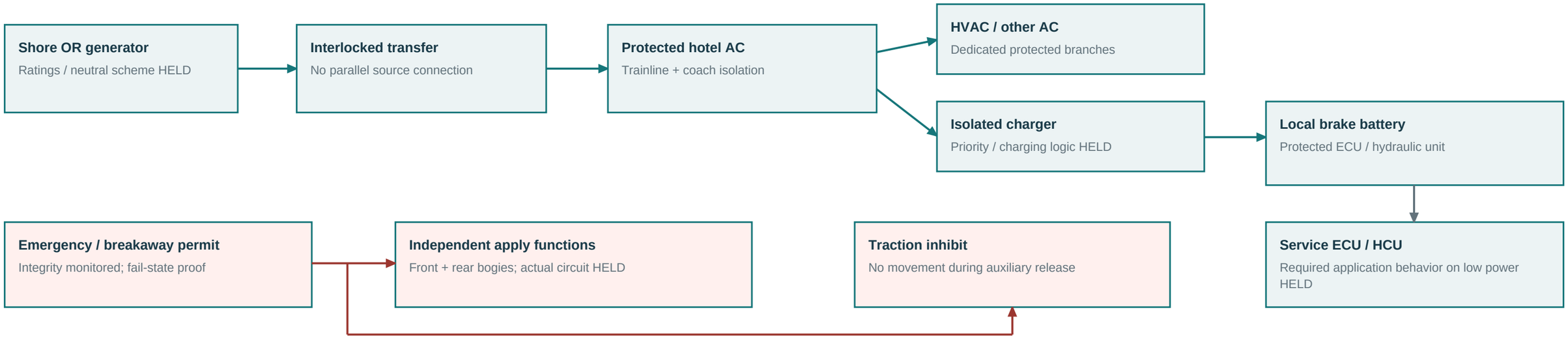
Wheel D	Regular axle	Accessible axle
400 mm	528.00 Nm	440.00 Nm
500 mm	660.00 Nm	550.00 Nm
596 mm	786.72 Nm	655.60 Nm

Diameters are sensitivity cases only.

596 mm reference is not a compatible bogie.

F=ma; axle torque=(m a/4) r. Actual axle load, braked-wheel count and worn radius control the final allocation.

H2 / HOTEL AND BRAKE POWER - FUNCTIONAL SEPARATION



The emergency path is independent of HVAC and an ordinary service PLC. No hydraulic hoses between vehicles in the preferred local-brake architecture. Power loss, open/short line, stalled controller, low battery, leakage and auxiliary-release faults require supplier-defined behavior and witnessed tests.

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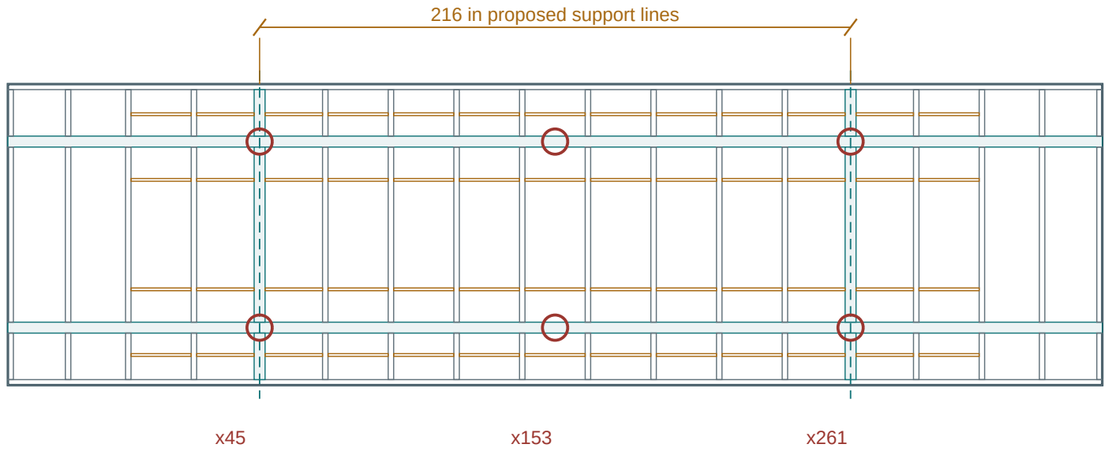
First article: measured evidence before design release

No passenger loading, public operation, pressure testing or rolling test is authorized by this sheet. Responsible specialists release load cases, restraints and test procedures.

V1 / INSTRUMENT THE LOAD PATH - PROPOSED LOCATIONS



Red rings: deflection / strain evidence zones; exact gauge position HELD.

Measure initial, under-load, unloading and residual geometry.

V2 / FIRST-ARTICLE TEST CONTRACT

Assembly	Required evidence
Frame	Load path, deflection, residual set, joint response
Floor	Actual span / seam / patch + moisture condition
Seat anchor	Actual feet, hard points, fasteners and load vector
Bogie fit	Tare/gross support height + all travel extrema
Brake	Force vs pressure; ramp, delay, empty/load balance
Power	Blackout, low battery, recharge and emergency path

Test load, cycles, duration, setup safety, instrument accuracy, pass/fail limits and release signatures remain assigned-engineer / OEM returns. Calculated screens do not supply acceptance criteria.

V3 / DRAWING RELEASE RETURN FIELDS

RUNNING GEAR

Rated load and spectrum; exact interface CAD; wheel / rail contract; support height and travel; bearing / brake / maintenance envelopes.

FRAME / BODY

Final member schedule; node and anchor calculations; weld / fastener specifications; surface treatment; tolerances; floor / seat qualification.

BRAKES / POWER

Actual force and release convention; hardware ratings; thermal / adhesion cases; fail-state diagram; battery / generator selection; tests and manuals.

Retain serial numbers, material certificates, calibration records, measured mass / centre of gravity, photographs, as-built changes and the disposition of every nonconformance. Repeat affected checks after a modification; do not erase the prior issue.

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